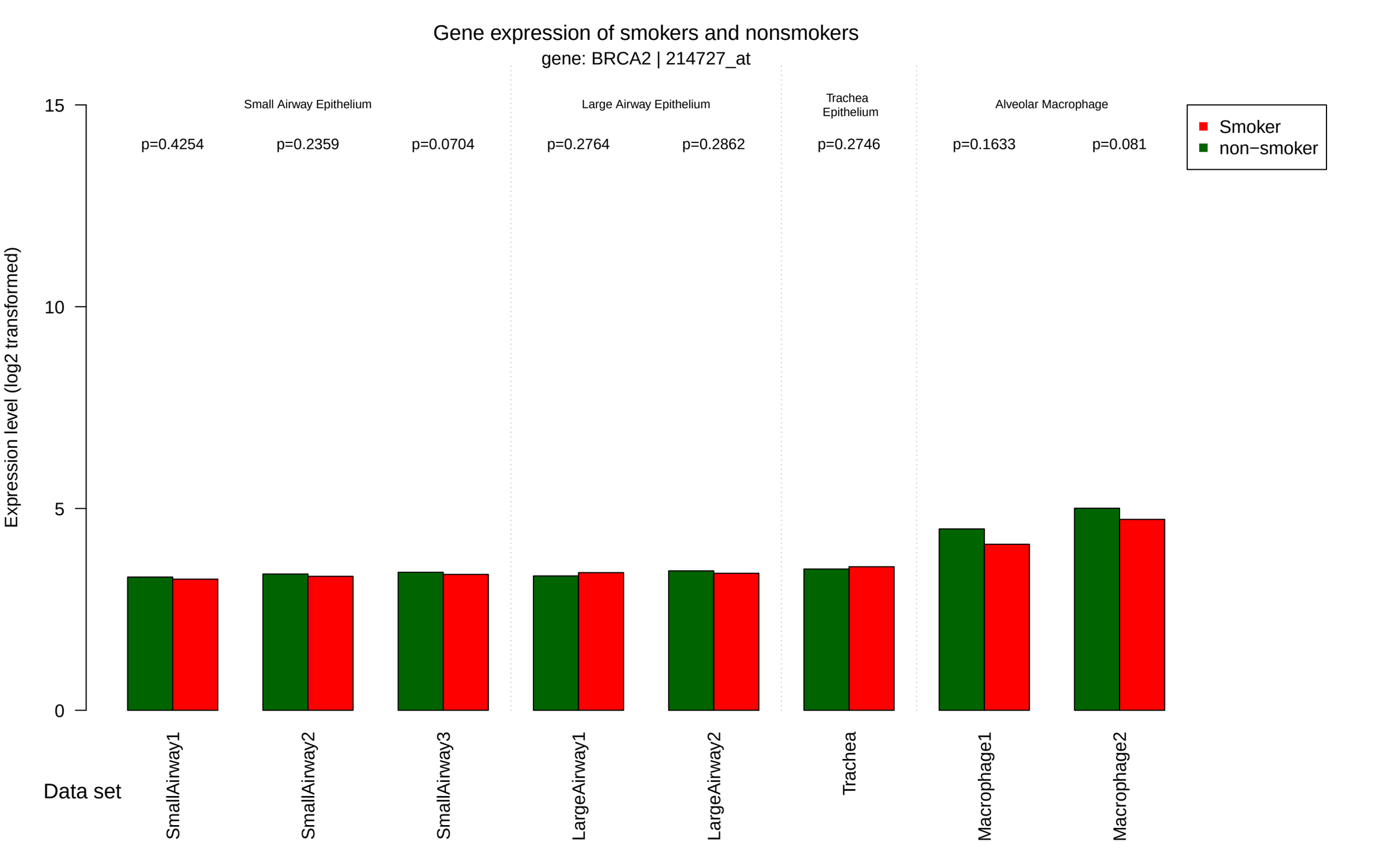
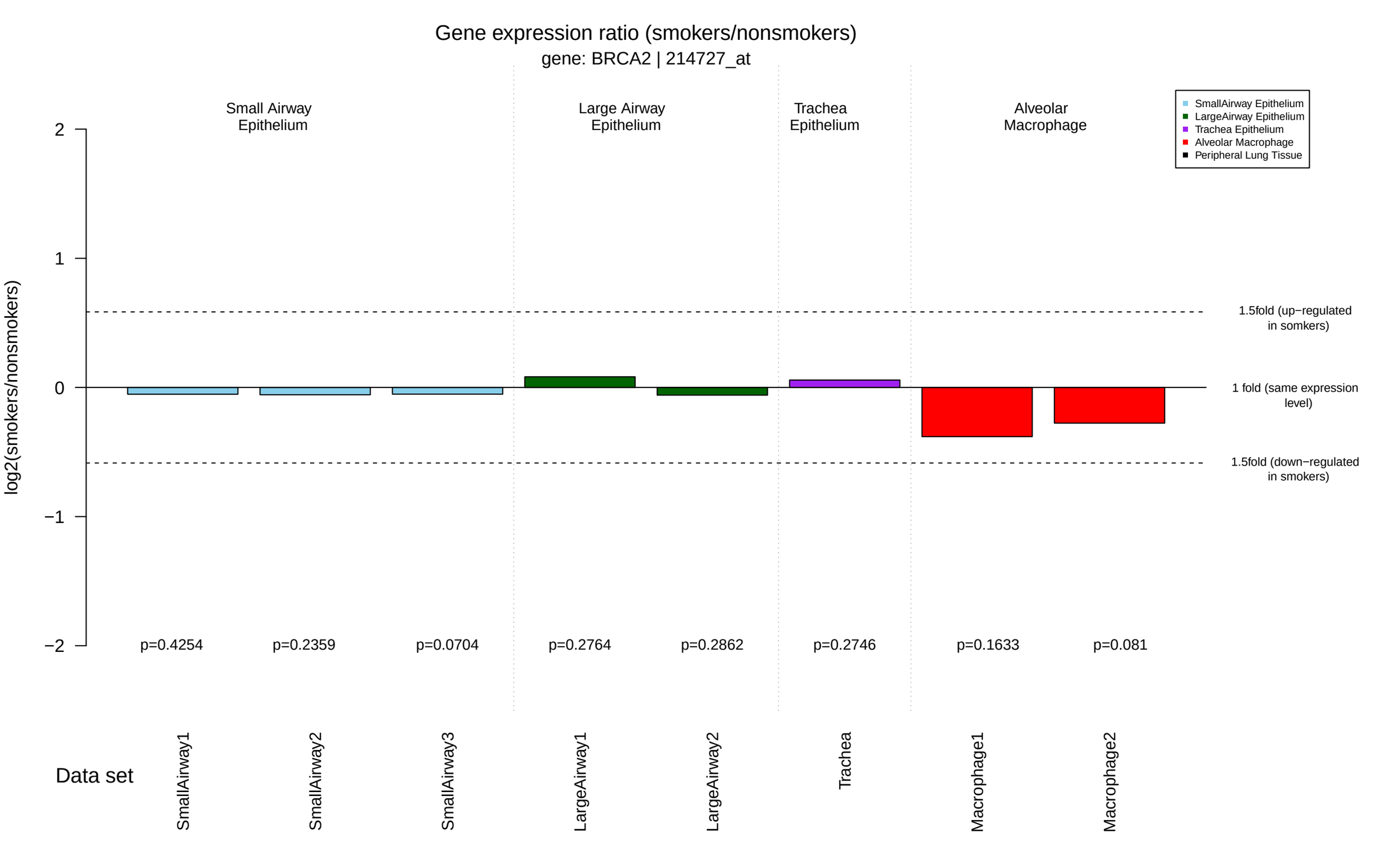
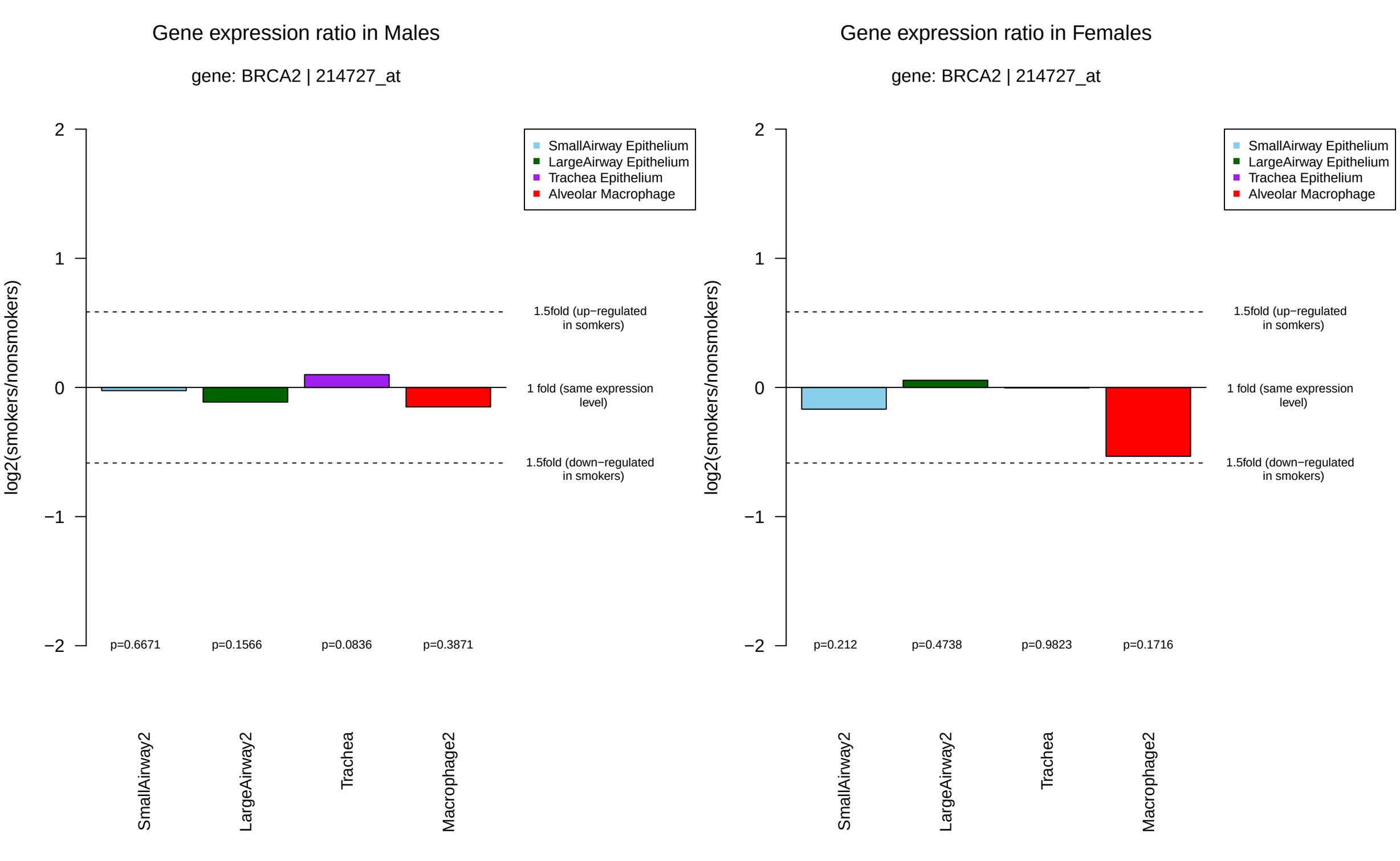
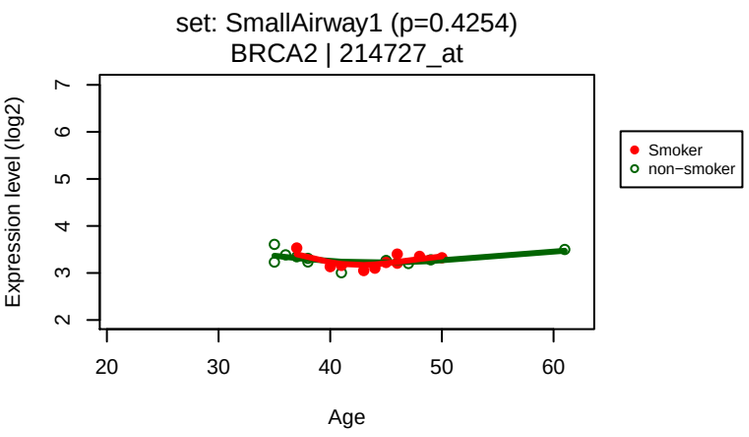
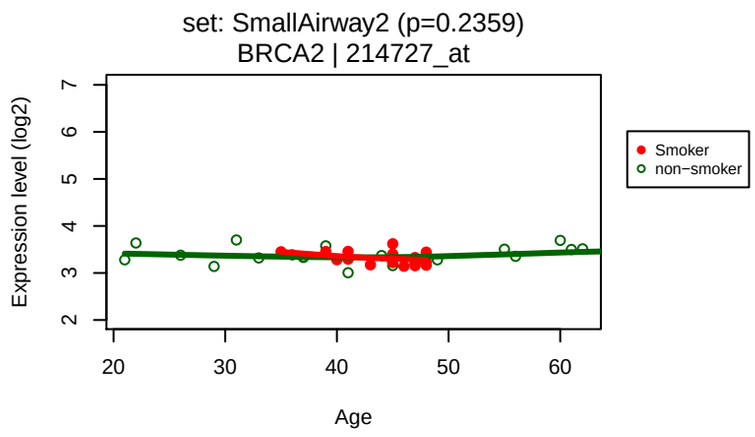
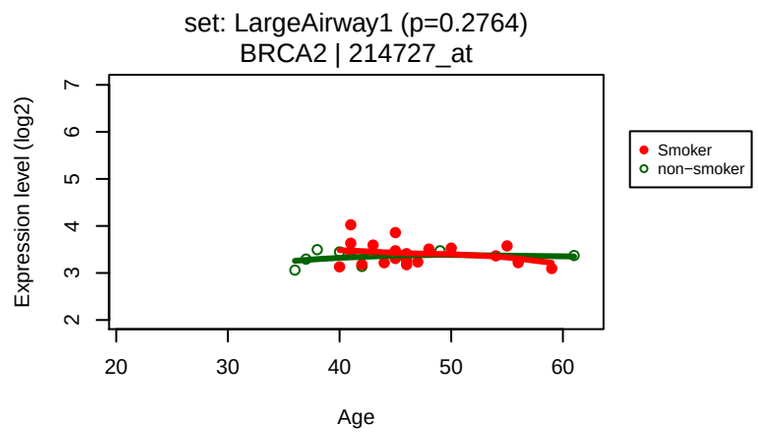

Gene expression of smokers and nonsmokers
gene: BRCA2 | 214727_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4254
p=0.2359
p=0.0704
p=0.2764
p=0.2862
p=0.2746
p=0.1633
p=0.081
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

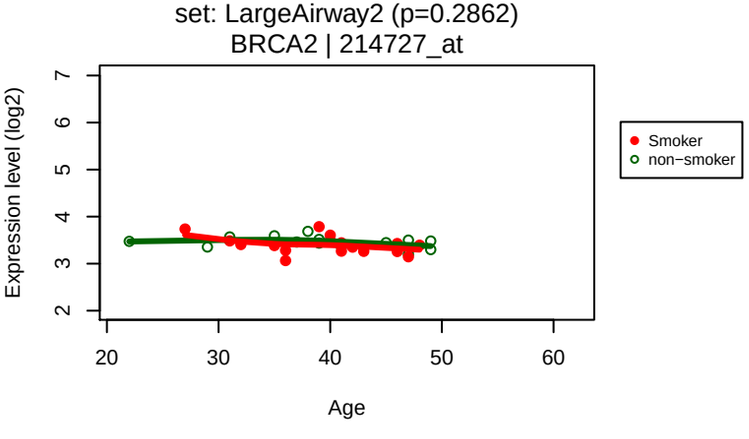

Gene expression ratio (smokers/nonsmokers)
gene: BRCA2 | 214727_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4254
p=0.2359
p=0.0704
p=0.2764
p=0.2862
p=0.2746
p=0.1633
p=0.081
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

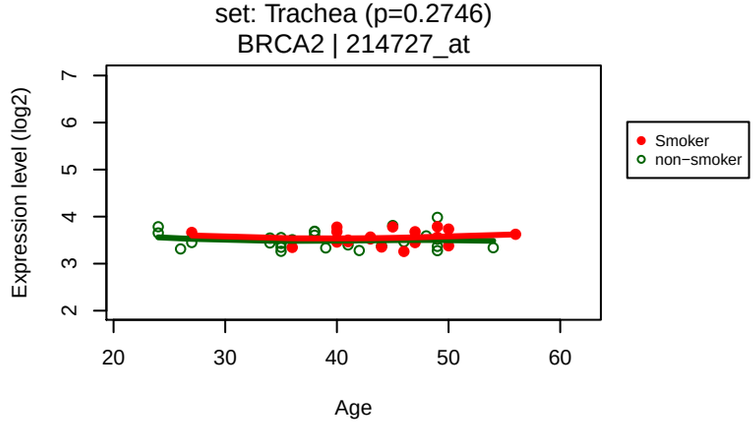

Gene expression ratio in Males
gene: BRCA2 | 214727_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6671
p=0.1566
p=0.0836
p=0.3871
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRCA2 | 214727_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.212
p=0.4738
p=0.9823
p=0.1716
SmallAirway2
LargeAirway2
Trachea
Macrophage2

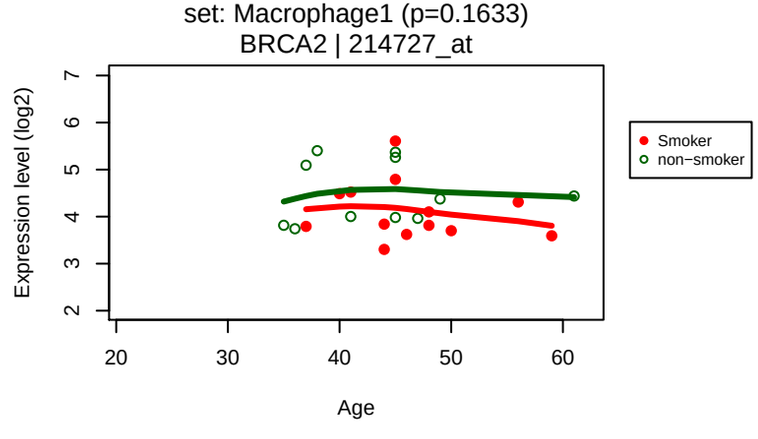

set: SmallAirway1 (p=0.4254)
BRCA2 | 214727_at
Expression level (log2)
Age
Smoker
non−smoker

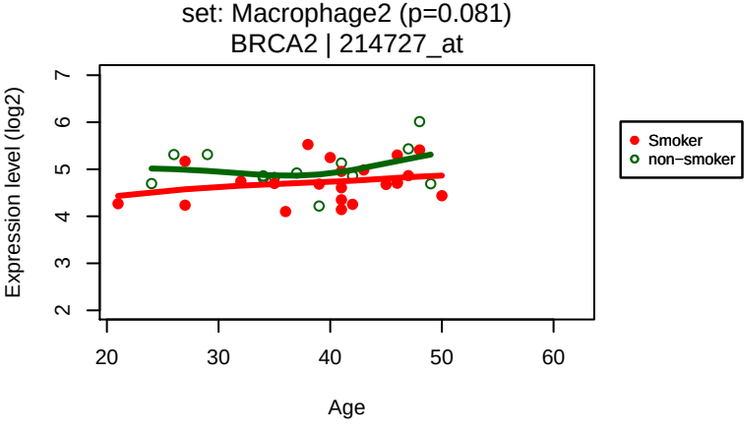

set: SmallAirway2 (p=0.2359)
BRCA2 | 214727_at
Expression level (log2)
Age
Smoker
non−smoker

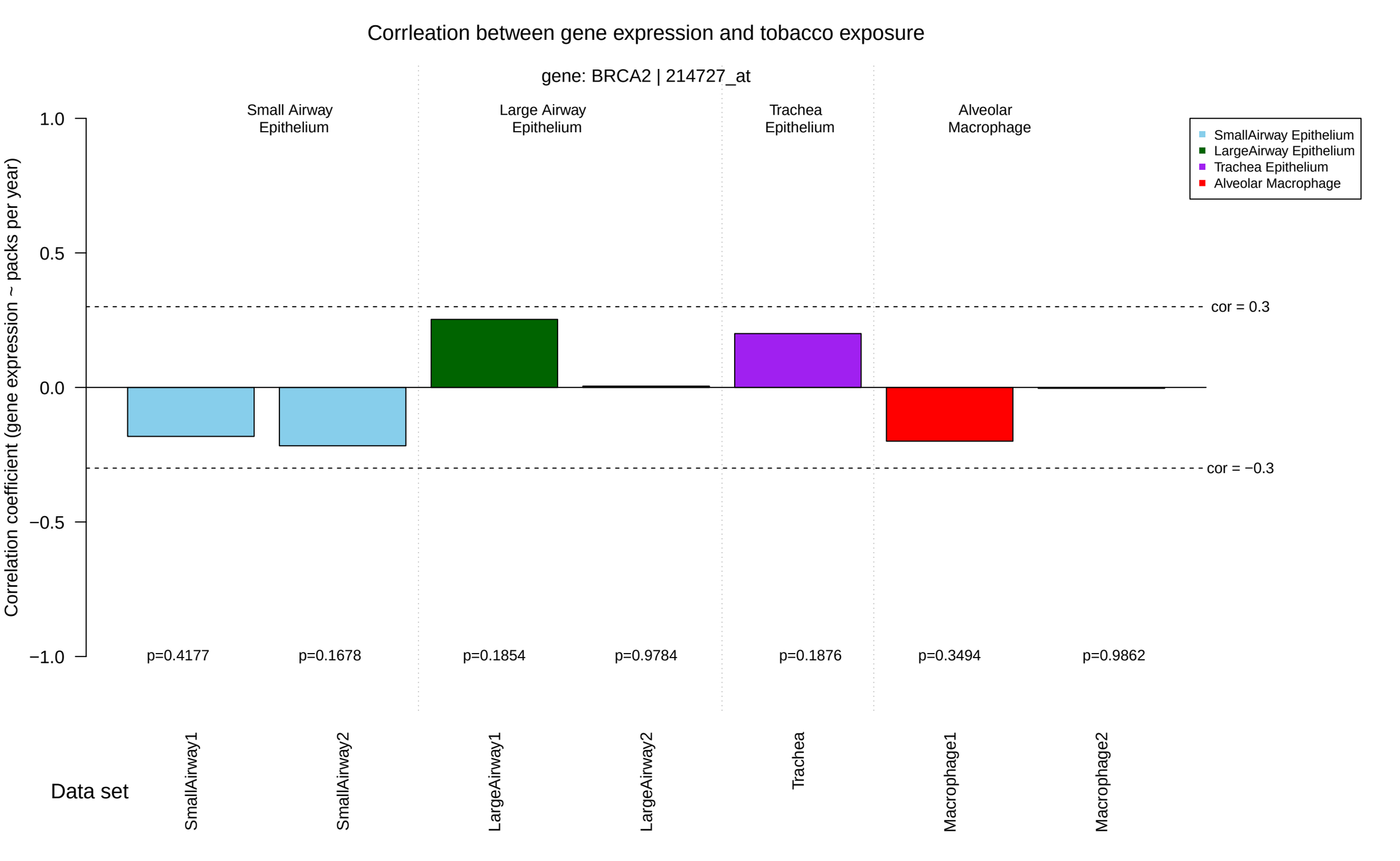

set: LargeAirway1 (p=0.2764)
BRCA2 | 214727_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2862)
BRCA2 | 214727_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2746)
BRCA2 | 214727_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1633)
BRCA2 | 214727_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.081)
BRCA2 | 214727_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRCA2 | 214727_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4177
p=0.1678
p=0.1854
p=0.9784
p=0.1876
p=0.3494
p=0.9862
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2